



WORK BREAKDOWN STRUCTURE (WBS) TEMPLATE

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WORK BREAKDOWN STRUCTURE (WBS)

DOST STARBOOKS: WHIZ CHALLENGE

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INTRODUCTION

The Work Breakdown Structure presented here represents all the work required to complete the STARBOOKS Whiz Challenge project — a gamified, offline-first educational platform developed for the Department of Science and Technology – Science and Technology Information Institute (DOST-STII). The system covers four interactive game modes, a badge and reward system, a player leaderboard, and a fully featured administrative dashboard, all deployed on DOST kiosk hardware via a LAN-based client-server architecture. The WBS is organized into five phases consistent with the project's lifecycle: Initiation, Planning, Executing, Monitoring and Controlling, and Closing. It adheres to the 100% Rule — every element of work required to produce the project deliverables is captured across these phases, with no work duplicated between branches and no work omitted. The WBS is decomposed to the work package level, where each package represents a discrete, assignable unit of work with an identifiable output, owner, and cost, tracked in OpenProject.

The WBS is broken down into the following views: Outline View, Hierarchical Structure, Tabular View, Tree Structure View (see the accompanying WBS charts), and WBS Dictionary. All budget figures are sourced from OpenProject and expressed in Philippine Peso (PHP).

OUTLINE VIEW

STARBOOKS Whiz Challenge

1. Initiation Phase

- a. Design Thinking Stage 1: Empathize
- b. PM Foundations Ch 2: Initiate a Project
 - i. Create Charter
 - ii. Create Objectives
 - iii. Create Scope
 - iv. Create Stakeholder Analysis
- c. Define Budgets
- d. Define Timelines

2. Planning Phase

- a. Design Thinking Stage 2: Define
- b. Design Thinking Stage 3: Ideate
- c. Requirements and Analysis Design Diagrams
 - i. Dataflow Diagrams
 1. Context Diagram
 2. Dataflow Diagram Level 1
 3. Dataflow Diagram Level 2 (Processes 2.1–2.8)



- ii. Use Cases
 - 1. Use Case Diagram
 - a. Fully Dressed Use Cases — Manage Player Account
 - b. Fully Dressed Use Cases — Play Game
 - c. Fully Dressed Use Cases — Reward Redemption
 - d. Fully Dressed Use Cases — Manage Admin Account
 - e. Fully Dressed Use Cases — Manage Quiz Settings
 - f. Fully Dressed Use Cases — Manage Player List
 - g. Fully Dressed Use Cases — View Analytics Dashboard
 - h. Fully Dressed Use Cases — View Leaderboard
 - iii. Entity Relationship Diagram
 - iv. Deployment Diagram
- d. Allocate Resources
- e. Subsidiary Management Plans
 - i. Plan Scope Management
 - ii. Plan Schedule Management
 - iii. Plan Quality Management
 - iv. Plan Resource Management
 - v. Plan Communications Management
 - vi. Plan Risk Management
 - vii. Plan Procurement Management
 - viii. Plan Stakeholder Engagement
 - ix. Plan Change Management
- 3. **Executing Phase**
 - a. Design Thinking Stage 4: Prototype
 - b. Design Thinking Stage 5: Test
 - c. Epic 01 — Application Development
 - i. Registration Page
 - ii. Player Login Page
 - iii. Homepage with Player Profile Section
 - iv. Player Leaderboard
 - v. Four Game Modes
 - 1. Whiz Challenge
 - 2. Whiz Memory Match
 - 3. Whiz Puzzle
 - 4. Whiz Battle
 - vi. Admin Dashboard
 - 1. Admin Login Page
 - 2. Analytics Dashboard
 - 3. Admin Leaderboard





- 4. List of Players
 - 5. List of Admins
 - 6. List of Questions
 - 7. Difficulty Level Settings
 - d. Sequence Diagrams
 - e. Deployment Diagram
 - f. Software Requirements Specification
- 4. **Monitoring and Controlling Phase**
 - a. Tracking Progress
 - b. Comparing Against the Plan
 - c. Identifying Deviations and Issues
 - d. Measuring Performance
- 5. **Closing Phase**
 - a. Completing All Tasks
 - i. User Acceptance Test Completions
 - b. Obtaining Final Approvals
 - c. Archiving Project Documentation
 - d. Conducting a Final Review
 - e. Project Turnover





HIERARCHICAL STRUCTURE

Level	WBS Code	Element Name
1	1	STARBOOKS Whiz Challenge
2	1.1	Initiation Phase
3	1.1.1	Design Thinking Stage 1: Empathize
3	1.1.2	PM Foundations Ch 2: Initiate a Project
4	1.1.2.1	Create Charter
4	1.1.2.2	Create Objectives
4	1.1.2.3	Create Scope
4	1.1.2.4	Create Stakeholder Analysis
3	1.1.3	Define Budgets
3	1.1.4	Define Timelines
2	1.2	Planning Phase
3	1.2.1	Design Thinking Stage 2: Define
3	1.2.2	Design Thinking Stage 3: Ideate
3	1.2.3	Requirements and Analysis Design Diagrams
4	1.2.3.1	Dataflow Diagrams
5	1.2.3.1.1	Context Diagram
5	1.2.3.1.2	Dataflow Diagram Level 1
5	1.2.3.1.3	Dataflow Diagram Level 2 (Processes 2.1–2.8)
4	1.2.3.2	Use Cases
5	1.2.3.2.1	Use Case Diagram
6	1.2.3.2.1.1	Fully Dressed Use Cases — Manage Player Account
6	1.2.3.2.1.2	Fully Dressed Use Cases — Play Game
6	1.2.3.2.1.3	Fully Dressed Use Cases — Reward Redemption
6	1.2.3.2.1.4	Fully Dressed Use Cases — Manage Admin Account
6	1.2.3.2.1.5	Fully Dressed Use Cases — Manage Quiz Settings
6	1.2.3.2.1.6	Fully Dressed Use Cases — Manage Player List
6	1.2.3.2.1.7	Fully Dressed Use Cases — View Analytics Dashboard
6	1.2.3.2.1.8	Fully Dressed Use Cases — View Leaderboard
4	1.2.3.3	Entity Relationship Diagram
4	1.2.3.4	Deployment Diagram
3	1.2.4	Allocate Resources
3	1.2.5	Subsidiary Management Plans
4	1.2.5.1	Plan Scope Management
4	1.2.5.2	Plan Schedule Management



TABULAR VIEW

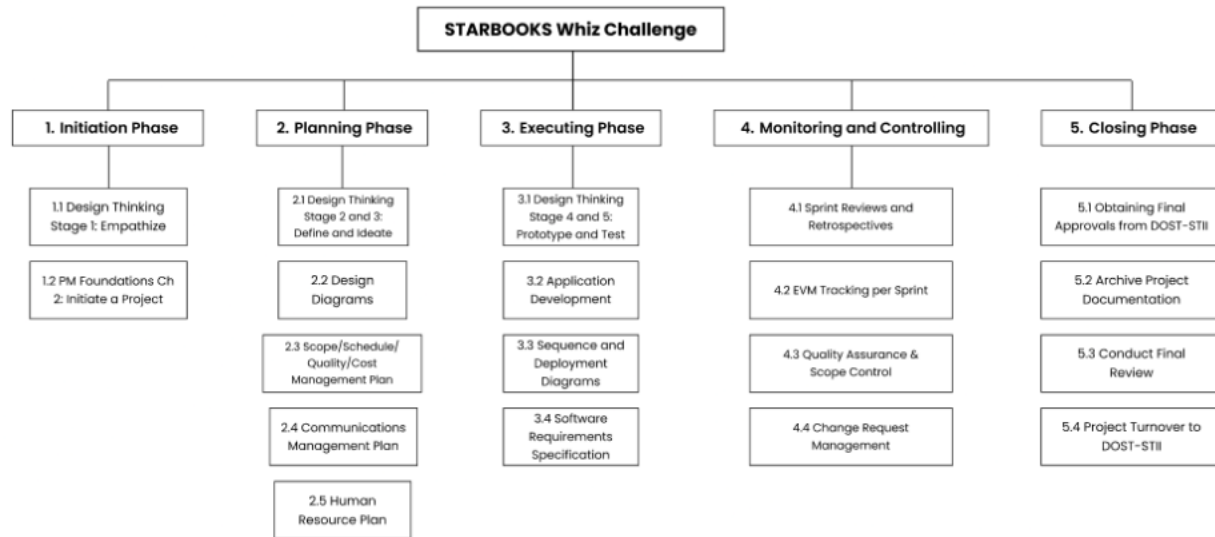
Level 1	Level 2	Level 3	Level 4
1. STARBOOKS Whiz Challenge	1.1 Initiation Phase	1.1.1 Design Thinking Stage 1: Empathize	—
		1.1.2 PM Foundations Ch 2: Initiate a Project	1.1.2.1 Create Charter
			1.1.2.2 Create Objectives
			1.1.2.3 Create Scope
			1.1.2.4 Create Stakeholder Analysis
			1.1.2.5 Milestone: PM Foundations Ch 2 Completed
		1.1.3 Define Budgets	—
		1.1.4 Define Timelines	—
	1.2 Planning Phase	1.2.1 Design Thinking Stage 2: Define	—
		1.2.2 Design Thinking Stage 3: Ideate	—
			1.2.3.1 Dataflow Diagrams (Context + Level 1 & 2)
			1.2.3.2 Use Cases (Use Case Diagram + 8 Fully Dressed Use Cases)
			1.2.3.3 Entity Relationship Diagram
			1.2.3.4 Deployment Diagram
		1.2.4 Allocate Resources	—
		1.2.5 Subsidiary Management Plans	1.2.5.1–1.2.5.9: Scope · Schedule · Quality · Resource · Communications · Risk · Procurement · Stakeholder · Change
	1.3 Executing Phase	1.3.1 Design Thinking Stage 4: Prototype	—



		1.3.2 Design Thinking Stage 5: Test	—
		1.3.3 Epic 01 — Application Development	1.3.3.1 Registration Page
			1.3.3.2 Player Login Page
			1.3.3.3 Homepage with Player Profile Section
			1.3.3.4 Player Leaderboard
			1.3.3.5 Four Game Modes (Whiz Challenge · Memory Match · Puzzle · Battle)
			1.3.3.6 Admin Dashboard (Login · Analytics · Leaderboard · Players · Admins · Questions · Difficulty Settings)
		1.3.4 Sequence Diagrams	—
		1.3.5 Deployment Diagram	—
		1.3.6 Software Requirements Specification	—
	1.4 Monitoring & Controlling	1.4.1 Tracking Progress	—
		1.4.2 Comparing Against the Plan	—
		1.4.3 Identifying Deviations and Issues	—
		1.4.4 Measuring Performance	—
	1.5 Closing Phase	1.5.1 Completing All Tasks	1.5.1.1 User Acceptance Test Completions
		1.5.2 Obtaining Final Approvals	—
		1.5.3 Archiving Project Documentation	—
		1.5.4 Conducting a Final Review	—
		1.5.5 Project Turnover	—



TREE STRUCTURE VIEW





Level	WBS Code	Element Name	Definition
1	1	STARBOOKS Whiz Challenge	All work required to design, develop, test, and deploy the STARBOOKS Whiz Challenge Version 1.0 — a gamified, offline-first STEM educational platform for DOST-STII.
2	1.1	Initiation Phase	All activities to formally start the project, including client discovery, PM Foundations initiation deliverables, budget definition, and timeline planning.
3	1.1.1	Design Thinking Stage 1: Empathize	Conduct empathy research with DOST-STII stakeholders to understand user needs, current system limitations, and the characteristics of the target student audience. Outputs feed directly into the Define and Ideate stages.
3	1.1.2	PM Foundations Ch 2: Initiate a Project	Complete all PM Foundations Chapter 2 deliverables required to formally initiate the project, including the Project Charter, Objectives, Scope, and Stakeholder Analysis.
4	1.1.2.1	Create Charter	Draft and finalize the Project Charter documenting the project's purpose, objectives, high-level scope, stakeholders, and Project Manager authorization.
4	1.1.2.2	Create Objectives	Define measurable project objectives aligned with DOST-STII's educational and operational goals for the STARBOOKS Whiz Challenge.
4	1.1.2.3	Create Scope	Develop the preliminary scope statement defining what is and is not included in the project.



4	1.1.2.4	Create Stakeholder Analysis	Identify all project stakeholders, assess their interests and influence, and document strategies for engagement.
4	1.1.2.5	Milestone: PM Foundations Ch 2 Completed	Formal milestone marking the completion and submission of all PM Foundations Chapter 2 initiation deliverables.
3	1.1.3	Define Budgets	Establish and document the phase-level budget allocations for all five project phases in OpenProject.
3	1.1.4	Define Timelines	Establish the project schedule and milestone dates across all phases, documented in OpenProject.
2	1.2	Planning Phase	All activities to produce the project's planning artifacts, including design thinking deliverables, all system analysis and design diagrams, resource allocation, and all subsidiary management plans.
3	1.2.1	Design Thinking Stage 2: Define	Synthesize empathy findings to define the problem statement and establish the project's design direction and goals.
3	1.2.2	Design Thinking Stage 3: Ideate	Brainstorm and evaluate candidate design solutions for the application, arriving at the selected design direction for development.
3	1.2.3	Requirements & Analysis Design Diagrams	Produce all system analysis and design diagrams required for the SRS, covering data flow, use cases, entity relationships, and deployment architecture.
4	1.2.3.1	Dataflow Diagrams	Develop the Context Diagram (Level 0), Dataflow Diagram Level 1, and all eight Level 2 DFDs covering each major system process.



4	1.2.3.2	Use Cases	Develop the Use Case Diagram and fully dressed use cases for all eight system functions, each including activity diagrams with swim lanes, sequence diagrams, and test cases.
4	1.2.3.3	Entity Relationship Diagram	Design and document the complete data model for the STARBOOKS Whiz Challenge system, covering all collections, fields, relationships, and constraints.
4	1.2.3.4	Deployment Diagram	Document the LAN-based client-server deployment architecture showing all hardware nodes (server PC, client PCs, LAN router) and software components (Laravel API, WebSocket server, MongoDB, Flutter web app).
3	1.2.4	Allocate Resources	Identify and assign all human and material resources required across project phases, ensuring adequate coverage for all scheduled work.
3	1.2.5	Subsidiary Management Plans	Develop all knowledge area management plans required to guide project execution through closing. Each plan defines the approach, tools, and processes for its respective knowledge area.
4	1.2.5.1	Plan Scope Management	Define how project scope will be defined, verified, and controlled, including the WBS and WBS Dictionary.
4	1.2.5.2	Plan Schedule Management	Define how the project schedule will be developed, maintained, and controlled.
4	1.2.5.3	Plan Quality Management	Define the quality standards, metrics, and control activities for all project deliverables.
4	1.2.5.4	Plan Resource Management	Define how team members will be identified, acquired, managed, and released throughout the project.



4	1.2.5.5	Plan Communications Management	Define the communication channels, frequency, formats, and responsibilities for all project stakeholders.
4	1.2.5.6	Plan Risk Management	Identify project risks, assess their probability and impact, and define response strategies.
4	1.2.5.7	Plan Procurement Management	Identify any procurement needs for tools, services, or resources, and define the procurement approach.
4	1.2.5.8	Plan Stakeholder Engagement	Define strategies to maintain productive engagement with DOST-STII, the faculty adviser, and other stakeholders.
4	1.2.5.9	Plan Change Management	Define the process for submitting, evaluating, approving, and implementing changes to the project scope, schedule, or budget.
2	1.3	Executing Phase	Full-scale development and testing of the STARBOOKS Whiz Challenge application, covering all player-facing features, the admin dashboard, and all supporting technical documentation.
3	1.3.1	Design Thinking Stage 4: Prototype	Develop prototypes of the application interfaces for validation with DOST-STII prior to full-scale development. Covers both player-side and admin-side UI prototypes.
3	1.3.2	Design Thinking Stage 5: Test	Conduct usability testing of prototypes with target users and DOST-STII stakeholders, collect feedback, and refine designs before development begins.



3	1.3.3	Epic 01 — Application Development	The complete development of all player-facing and admin-facing features of the STARBOOKS Whiz Challenge application, built iteratively across sprints using Flutter/Dart (frontend) and Laravel/MongoDB (backend).
4	1.3.3.1	Registration Page	Develop the multi-step player registration interface collecting username, password, school, age, avatar, category, sex, region, province, and city. Includes form validation, API integration, password hashing, and success/error messaging.
4	1.3.3.2	Player Login Page	Develop the player authentication interface with username/password entry, credential validation against the database, session token management, and appropriate error messaging.
4	1.3.3.3	Homepage with Player Profile Section	Develop the main homepage displaying the player's avatar, username, earned badges, and navigation buttons to all four game modes. Includes dynamic data fetching from the backend.
4	1.3.3.4	Player Leaderboard	Develop the leaderboard screen showing top player rankings with username, avatar, total rewards, reward breakdowns by difficulty, and toggle between Whiz Challenge and Whiz Battle views.
4	1.3.3.5	Four Game Modes	Develop all four interactive game modes of the STARBOOKS Whiz Challenge, each supporting Science and Math categories and three difficulty levels (Easy, Average, Difficult).



5	1.3.3.5.1	Whiz Challenge	Develop the solo quiz game mode with category and difficulty selection, randomized question loading, countdown timer, answer validation with visual feedback, score calculation, and badge award logic for perfect scores.
5	1.3.3.5.2	Whiz Memory Match	Develop the card-matching game with randomized grid generation by difficulty, card flip logic, pair matching detection, elapsed timer, move counter, star scoring, and fastest time tracking.
5	1.3.3.5.3	Whiz Puzzle	Develop the jigsaw puzzle game with image selection by category and difficulty, drag-and-drop piece mechanics, proximity-based snapping, elapsed timer, move counter, completion detection, and fastest time tracking.
5	1.3.3.5.4	Whiz Battle	Develop the real-time 1v1 quiz battle mode with game code generation, WebSocket-based player matchmaking, synchronized question delivery, per-question score updates, winner determination, and badge award for the winner.
4	1.3.3.6	Admin Dashboard	Develop the complete web-based administrative dashboard for DOST-STII staff, covering authentication, analytics, leaderboard management, player management, admin management, content management, and game configuration.
5	1.3.3.6.1	Admin Login Page	Develop the admin-specific authentication interface accessible via a dedicated admin link, with credential validation, session management, and redirect to the analytics dashboard on success.



5	1.3.3.6.2	Analytics Dashboard	Develop the analytics dashboard displaying key metrics including total registered players, most played game mode, average session duration, gender and age distribution, regional breakdown, game mode preferences by gender, and reward distribution. Includes export functionality.
5	1.3.3.6.3	Admin Leaderboard	Develop the admin view of the leaderboard showing top player rankings with badge and star breakdowns, toggleable by game mode, with export functionality for DOST-STII record-keeping.
5	1.3.3.6.4	List of Players	Develop the player management interface allowing admins to view all registered players, add new players, edit profiles, reset passwords, view badge collections, process reward claims, and delete accounts.
5	1.3.3.6.5	List of Admins	Develop the admin management interface allowing authorized admins to view, add, edit, change passwords for, and delete other administrator accounts. Includes role-based access enforcement preventing self-deletion.
5	1.3.3.6.6	List of Questions	Develop the question bank management interface allowing admins to view, search, filter, add, edit, and delete quiz questions by category (Science/Math), difficulty level (Easy/Average/Difficult), and status.
5	1.3.3.6.7	Difficulty Level Settings	Develop the game configuration interface allowing admins to set the number of questions per session and time limit per question for each difficulty level, with validation and immediate effect on live game modes.



3	1.3.4	Sequence Diagrams	Develop all sequence diagrams for the SRS, illustrating the message flow between actors and system components for all use case scenarios.
3	1.3.5	Deployment Diagram	Finalize the deployment diagram reflecting the Docker-based production architecture, including all container definitions and LAN network topology.
3	1.3.6	Software Requirements Specification	Compile and finalize the complete SRS document covering all functional and non-functional requirements, UML diagrams, test cases, and group logs for submission to DOST-STII and the faculty adviser.
2	1.4	Monitoring and Controlling Phase	Ongoing oversight activities conducted throughout the project lifecycle to track performance, compare actual progress against the approved plan, identify deviations, and take corrective action as needed.
3	1.4.1	Tracking Progress	Monitor the status of all work packages in OpenProject at the end of each sprint, recording actual hours, costs, and task completion against the plan.
3	1.4.2	Comparing Against the Plan	At the end of each sprint, calculate EVM metrics (PV, EV, AC, SV, CV, SPI, CPI) and compare actual performance against the approved scope, schedule, and cost baselines.
3	1.4.3	Identifying Deviations and Issues	Identify variances, scope gaps, risks, and blockers through sprint reviews and retrospectives, and document findings for corrective action.



3	1.4.4	Measuring Performance	Assess overall project health against the cost and schedule performance index thresholds defined in the Cost Management Plan, and report findings to the Project Sponsor.
2	1.5	Closing Phase	All activities required to formally close the STARBOOKS Whiz Challenge project, including completing outstanding work, obtaining DOST-STII acceptance, archiving documentation, and transferring the complete system to DOST-STII.
3	1.5.1	Completing All Tasks	Ensure all open work packages, bug fixes, and outstanding documentation items are completed and verified prior to formal project closure.
4	1.5.1.1	User Acceptance Test Completions	Conduct final user acceptance testing with DOST-STII on the deployed system, validate all functional and non-functional requirements, and obtain sign-off on test results.
3	1.5.2	Obtaining Final Approvals	Present the completed system to DOST-STII for formal acceptance review. The Project Sponsor signs the project acceptance document confirming that all deliverables meet agreed requirements.
3	1.5.3	Archiving Project Documentation	Compile and archive all project documents including the SRS, SDD, Scope Management Plan, Cost Management Plan, WBS, meeting minutes, test cases, and source code repository.
3	1.5.4	Conducting a Final Review	Conduct a final sprint retrospective and lessons learned session with the full project team, documenting what went well, what could be improved, and recommendations for future similar projects.



3	1.5.5	Project Turnover	Formally transfer the complete system — including source code, database, deployment package, user manuals, setup instructions, and all project documentation — to DOST-STII for their ownership and future maintenance.
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GLOSSARY OF TERMS

Level of Effort (LOE): The amount of work required to complete a task, typically expressed in hours. All labor in this project is rated at PHP 500.00 per hour per team member.

WBS Code: A unique identifier assigned to each element in the Work Breakdown Structure to designate its hierarchical location. In this document, WBS codes follow the OpenProject numbering convention (e.g., 3.3.5.1 for Whiz Challenge).

Work Package: A deliverable or work component at the lowest level of a WBS branch. In this project, work packages are the individual features, tasks, and summary tasks tracked in OpenProject, each with an assigned owner, budget, and completion status.

WBS Component: Any component of a WBS located at any level — whether a Work Package or a WBS Element containing other elements.

WBS Element: A single WBS component located anywhere within the WBS. A WBS Element may contain work directly or may contain other WBS Elements or Work Packages below it.

100% Rule: The principle that the WBS must capture 100% of the work defined in the project scope — no more and no less. Every deliverable and work effort required to complete the STARBOOKS Whiz Challenge is represented somewhere in this WBS, and no work is duplicated across branches.

Milestone: A significant event or achievement in the project with zero duration. In this WBS, the completion of PM Foundations Chapter 2 and User Acceptance Test Completions are formally designated milestones.



- Epic:** In the Agile/Scrum methodology used for this project, an Epic is a large body of work that can be broken down into features and user stories. Epic 01 (WBS 1.3.3) encompasses the full application development effort for the STARBOOKS Whiz Challenge.
- Feature:** A distinct functional capability of the system delivered within an Epic. Features in this WBS (e.g., Registration Page, Whiz Challenge, Admin Dashboard) correspond directly to deliverable increments produced within sprints.

